ACS DIVISION OF CHEMICAL EDUCATION EXAMINATION

Form 2018

GENERAL-ORGANIC-BIOCHEMISTRY

Serial No. 05625

Prepared by the Examinations Institute of the American Chemical Society Division of Chemical Education through its General-Organic-Biochemistry Committee.

General-Organic-Biochemistry Committee

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TO THE EXAMINER:

This test is designed to be taken with a special answer sheet on which the student records his or her responses. All answers are to be marked on this answer sheet, not in the test booklet. Each student should be provided with a test booklet, one answer sheet, and scratch paper; all of which must be turned in at the end of the examination period. The test is to be available to the students only during the examination period. You must collect and account for all exam booklets at the end of the examination. For complete instructions refer to the Directions for Administering Examinations. Only nonprogrammable calculators are permitted. ***All electronic devices with photo-taking capability are prohibited.*** Norms are based on:

Subtest Part	I. General	II. Organic	III. Biochemistry
Number of items	50	50	50
Time limit, minutes	60	60	60

TO THE STUDENT:

DO NOT WRITE ANYTHING IN THIS BOOKLET! Do not turn this page until your instructor gives the signal to begin. A periodic table and datasheet are provided on page 16. When you are told to begin work, open the booklet and read the directions on page 2.

Stock Code GB18

A C S
Exams

DIRECTIONS

- When you have selected your answer, blacken the corresponding space on the answer sheet with a soft, black #2 pencil. Make a heavy, full mark, but no stray marks. If you decide to change an answer, erase the unwanted mark very carefully.
- Make no marks in the test booklet. Do all calculations on scratch paper provided by your instructor.
- There is only one correct answer to each question. Any questions for which more than one response has been blackened **will not be counted**.
- Your score is based solely on the number of questions you answer correctly. **It is to your advantage to answer every question.**
- Your test, answer sheet, and scratch paper must be turned in before leaving the examination room.

GENERAL CHEMISTRY SECTION

1. Which statement best represents a chemical change of aluminum?

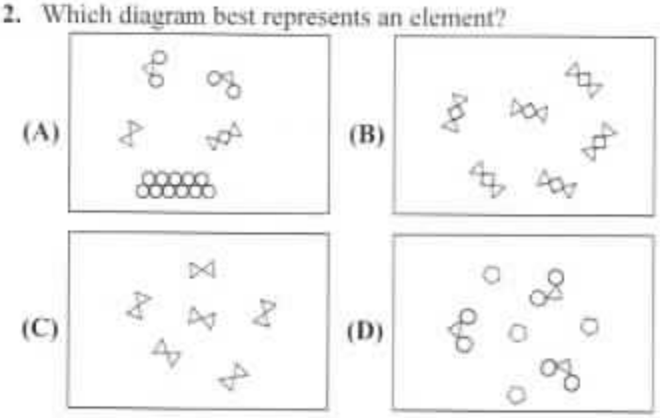
(A) Aluminum metal melts at 630.3 °C.

(B) Aluminum is a silvery white metal.

(C) The density of aluminum metal is 2.70 $g{\cdot}cm^{-3}$.

(D) Aluminum metal reacts with oxygen gas to form aluminum oxide.

2. Which diagram best represents an element?



3. Which state(s) of matter can be compressed significantly?

(A) gas **(B)** liquid

(C) solid **(D)** gas and liquid

4. Air is an example of a(n)

(A) compound. **(B)** element.

(C) heterogenous mixture. **(D)** homogeneous mixture.

5. The pediatric daily dosage of ampicillin is 150 $mg{\cdot}kg^{-1}$. What mass (in mg) of ampicillin is needed for a 25 lb child per day? (1 kg = 2.20 lb)

(A) 1.7 mg **(B)** 11 mg

(C) 13 mg **(D)** 1700 mg

6. What is smaller than 2.5 mL?

(A) 2.5×10^{6} nL **(B)** 250 µL

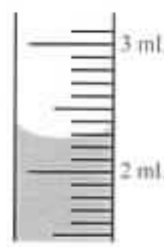
(C) 25 dL **(D)** 0.25 L

7. Diamond has a density of 3.52 $g{\cdot}mL^{-1}$. What is the volume (in mL) of a 75-carat diamond (1 carat = 0.200 g)?

(A) 53 mL **(B)** 4.3 mL

(C) 1.3 mL **(D)** 0.23 mL

8. A laboratory technician is tasked to take a volume reading from the graduated cylinder shown. Which reported measurement is most correct?



(A) 2.00 mL **(B)** 2.3 mL

(C) 2.28 mL **(D)** 3.75 mL

9. Which elements have the same number of neutrons?

I	$^{28}_{15}P$
II	$^{24}_{12}Mg$
III	$^{25}_{12}Mg$
IV	$^{28}_{14}Si$

(A) I and II **(B)** I and III

(C) II and IV **(D)** I, II and III

10. What is the ground state electronic configuration of the Ca atom?

(A) $1s^2\,2s^2\,2p^6\,3s^2$

(B) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6$

(C) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2$

(D) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,4p^6\,5s^1$

11. Which ions have the same total number of electrons?

(A) Ca^{2+} and Mg^{2+} **(B)** Ca^{2+} and Cl^-
(C) Ca^{2+} and Cr^{3+} **(D)** Cl^- and Cr^{3+}

12. Which element forms two or more ions with different ionic charges?

(A) Ca **(B)** F **(C)** Fe **(D)** K

13. Which compound is best described as only containing ionic bonds?

(A) H_2 **(B)** H_2O
(C) NH_4Cl **(D)** RbF

14. Which compound contains a polyatomic ion with a 3– charge?

(A) $CaCO_3$ **(B)** Li_2SO_4
(C) Mg_3N_2 **(D)** Na_3PO_4

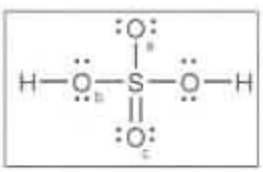
15. What is the chemical formula for the substance formed when strontium and oxygen combine?

(A) SrO **(B)** Sr_2O **(C)** SrO_2 **(D)** Sr_2O_2

16. What is the chemical formula for sodium bicarbonate, used in the maintenance of blood pH?

(A) $NaCO_3$ **(B)** Na_2CO_3
(C) $NaHCO_3$ **(D)** $Na(HCO_3)_2$

17. The Lewis structure of sulfuric acid (H_2SO_4) is incorrect. Which structural feature needs to be changed to give a correct Lewis structure?

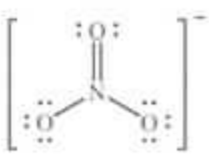


(A) add a bond between S and O_a
(B) add a bond between S and O_b
(C) remove a lone pair on O_a
(D) add a lone pair on O_c

18. In the Lewis structure of NH_2^-, the nitrogen atom makes two single bonds to the hydrogen atoms and has two lone pairs of electrons. What is the H–N–H bond angle in NH_2^-?

(A) 90° **(B)** 104° **(C)** 120° **(D)** 180°

19. How many total possible resonance structures are there for the nitrate ion, NO_3^-?



(A) 4 **(B)** 3 **(C)** 2 **(D)** 1

20. What is the pH of a sample with a $[OH^-] = 2.6\times10^{-6}$ at 25 °C?

(A) 5.58 **(B)** 6.22 **(C)** 8.41 **(D)** 10.41

21. Ammonium carbonate is commonly referred to as a "smelling salt" because it is used to revive people who have fainted. What is the molar mass of $(NH_4)_2CO_3$?

(A) 76.0 $g{\cdot}mol^{-1}$ **(B)** 82.1 $g{\cdot}mol^{-1}$
(C) 96.1 $g{\cdot}mol^{-1}$ **(D)** 120.1 $g{\cdot}mol^{-1}$

22. What mass of NO is formed when 0.640 g of molecular nitrogen undergoes this reaction? $N_2(g) + O_2(g) \rightarrow 2NO(g)$

Molar mass / $g{\cdot}mol^{-1}$	
N_2	28.02
NO	30.01

(A) 0.300 g **(B)** 0.680 g
(C) 1.19 g **(D)** 1.37 g

23. If 13.2 mol $AlCl_3$ are produced when 10.0 mol Al_2O_3 reacts with an excess amount of HCl, what is the percent yield?

$$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$$

(A) 22.0% **(B)** 66.0% **(C)** 100.% **(D)** 132%

24. When 10.0 g of NH_3 reacts with 10.0 g NO to yield N_2 and water, what is the limiting reagent?

$$4NH_3(g) + 6NO(g) \rightarrow 5N_2(g) + 6H_2O(l)$$

Molar mass / $g{\cdot}mol^{-1}$	
NH_3	17.04
NO	30.00
N_2	28.02
H_2O	18.02

(A) H_2O **(B)** NH_3 **(C)** N_2 **(D)** NO

25. Which statement is NOT part of the kinetic molecular theory of gases?

(A) Gas particles move rapidly.
(B) Gas particles do not attract or repel one another.
(C) Gas particles have very little empty space between them.
(D) Gas particles move faster when the temperature increases.

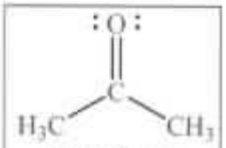
26. Which physical state has the weakest intermolecular forces?

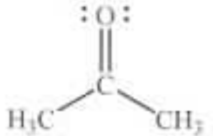
(A) gas **(B)** liquid **(C)** plasma **(D)** solid

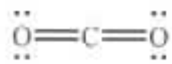
27. If a 2.50 L oxygen tank has a pressure of 3.45 atm at 25.0 °C, how many moles of gas are present in the container?

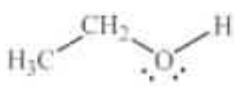
(A) 0.353 mol **(B)** 0.845 mol
(C) 2.84 mol **(D)** 211 mol

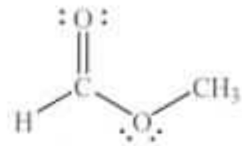
28. Which molecule will form hydrogen bonds with propanone?


propanone

(A) 

(B) 

(C) 

(D) 

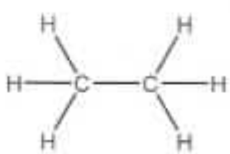
29. What volume of 1.50 M anesthetic is required to prepare 130. mL of 0.244 M anesthetic solution?

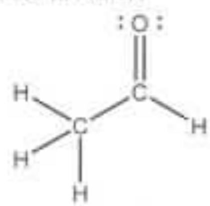
(A) 2.83 mL **(B)** 5.18 mL
(C) 21.1 mL **(D)** 799 mL

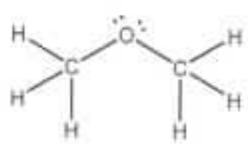
30. Normal saline solution is 0.920 % (m/v) NaCl in water. What mass (in g) of NaCl would be dissolved in water to prepare 2.50 L of saline?

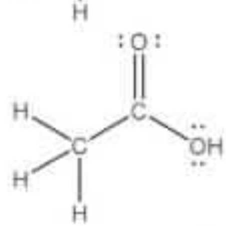
(A) 23.0 g **(B)** 26.0 g **(C)** 53.8 g **(D)** 58.8 g

31. Which compound is the most soluble in water?

(A) 

(B) 

(C) 

(D) 

32. In osmosis, only ________ moves through a semipermeable membrane into an area of ________ concentration.

(A) solute; lower solute
(B) solute; higher solute
(C) solvent; lower solute
(D) solvent; higher solute

33. The human body and a car both combust fuel. What do both combustion reactions yield?

(A) CO_2 **(B)** N_2O **(C)** O_2 **(D)** SO_2

34. A student drops a chunk of solid into a clear, colorless liquid. The solid disappears slowly, and bubbles are generated after the addition of the solid. Which chemical equation best fits the series of events?

(A) $I_2(s) + 2KBr(aq) \rightarrow 2KI(aq) + Br_2(l)$
(B) $Zn(s) + 2HCl(aq) \rightarrow H_2(g) + ZnCl_2(aq)$
(C) $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$
(D) $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

35. What is an example of a double displacement reaction?

(A) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
(B) $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$
(C) $2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + Al_2O_3(s)$
(D) $BaCl_2(aq) + 2KOH(aq) \rightarrow 2KCl(aq) + Ba(OH)_2(s)$

36. What are the products when Fe(s) and $CuSO_4(s)$ react?

(A) $CuO(s) + FeS(s)$
(B) $Cu(s) + FeSO_4(s)$
(C) $Cu(s) + Fe(s) + SO_4(s)$
(D) $Cu(s) + FeS(s) + O_2(g)$

37. What is the pH of a sample with a $[H^+] = 4.6\times10^{-6}$ M?

(A) 1.59 **(B)** 5.34 **(C)** 7.78 **(D)** 10.55

38. Which reaction is an acid/base reaction?

(A) $2NH_4NO_3(s) \rightarrow N_2(g) + O_2(g) + 4H_2O(l)$
(B) $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
(C) $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
(D) $H_3PO_4(aq) + 3LiOH(aq) \rightarrow Li_3PO_4(aq) + 3H_2O(l)$

39. What are the products when HCl(aq) and $Ba(OH)_2(aq)$ react?

(A) $H_2O_2(aq) + BaCl_2(aq)$ **(B)** $BaCl(aq) + H_2O(l)$
(C) $BaCl_2(aq) + H_2O(l)$ **(D)** $BaH_2(aq) + ClO(aq)$

40. What is the conjugate base in the reaction below?
$CH_3NH_2 + CH_3COOH \rightarrow CH_3NH_3^+ + CH_3COO^-$

(A) CH_3NH_2 **(B)** CH_3COOH
(C) $CH_3NH_3^+$ **(D)** CH_3COO^-

41. If working with X-ray radiation in a hospital setting, what is the minimum protection a health care worker should take?

(A) wear a lead vest

(B) wear aluminum foil

(C) wear gloves and lab coat

(D) no special measures needed

42. Gallium-68, which has a half-life of 68 minutes, is used to detect pancreatic cancer. If a patient were given 5.0 mg of gallium-68 at 8:00 am, at what approximate time would the patient have only 0.15 mg of gallium-68 inside them?

(A) 12:00 pm **(B)** 2:00 pm

(C) 7:00 pm **(D)** 10:00 pm

43. What product is missing in the nuclear decay?

$$^{232}_{90}\mathrm{Th} \rightarrow\ ?\ + {}^{4}_{2}\mathrm{He}$$

(A) $^{228}_{88}\mathrm{Ra}$ **(B)** $^{232}_{88}\mathrm{Ra}$ **(C)** $^{228}_{90}\mathrm{Ra}$ **(D)** $^{232}_{90}\mathrm{Ra}$

44. Polonium-210 is an alpha emitter with a half-life of around 138 days. What is the final stable isotope?

(A) $^{214}\mathrm{Bi}$ **(B)** $^{208}\mathrm{Hg}$ **(C)** $^{206}\mathrm{Pb}$ **(D)** $^{214}\mathrm{Rn}$

45. What is the equilibrium constant expression for the reaction?

$$\mathrm{Co(H_2O)_6^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons CoCl_4^{2-}(aq) + 6H_2O(l)}$$

(A) $K = \dfrac{\left[\mathrm{CoCl_4^{2-}}\right]}{\left[\mathrm{Co(H_2O)_6^{2+}}\right]}$

(B) $K = \left[\mathrm{Co(H_2O)_6^{2+}}\right]\left[\mathrm{Cl^-}\right]^4$

(C) $K = \dfrac{\left[\mathrm{Co(H_2O)_6^{2+}}\right]\left[\mathrm{Cl^-}\right]^4}{\left[\mathrm{CoCl_4^{2-}}\right]}$

(D) $K = \dfrac{\left[\mathrm{CoCl_4^{2-}}\right]}{\left[\mathrm{Co(H_2O)_6^{2+}}\right]\left[\mathrm{Cl^-}\right]^4}$

46. Consider the reaction

$$\mathrm{A(g) + B(g) \rightleftharpoons C(g) + D(g)}$$

with an equilibrium constant of 3.2×10^5. At equilibrium, the reaction would consist of ____.

(A) mostly reactants

(B) mostly products

(C) fewer products

(D) equal quantities of both

47. What would happen to the equilibrium below if the pressure was increased due to a change in volume?

$$\mathrm{2H_2O(g) + N_2(g) \rightleftharpoons 2H_2(g) + 2NO(g)}$$

(A) shifts to the left **(B)** shifts to the right

(C) K will increase **(D)** K will decrease

48. Which statement is TRUE for an endothermic reaction?

(A) Increasing the reaction temperature increases the value of the equilibrium constant K.

(B) Increasing the reaction temperature decreases the value of the equilibrium constant K.

(C) Increasing the reaction temperature has no effect on the value of the equilibrium constant K.

(D) Not enough information is available to determine the effect of increasing the reaction temperature on the equilibrium constant K.

49. To increase the rate of a reaction in the forward direction, which variable would you need to increase?

(A) pressure

(B) temperature

(C) container volume

(D) product concentration

50. What best describes the reaction?

$$\mathrm{NH_4NO_3(s) + energy \rightleftharpoons NH_4^+(aq) + NO_3^-(aq)}$$

(A) endothermic **(B)** exothermic

(C) isothermic **(D)** mesothermic

End of General Chemistry Section

ORGANIC CHEMISTRY SECTION

ACS **51.** How is a saturated organic compound best described?

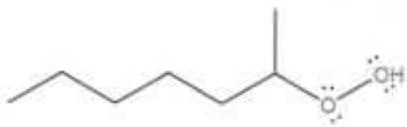
(A) contains only one carbon-carbon bond

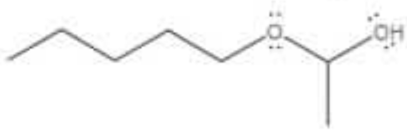
(B) contains only carbon-carbon single bonds

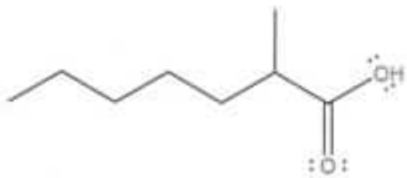
(C) contains carbon-carbon double bonds

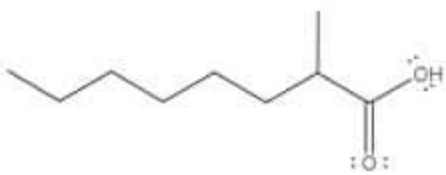
(D) contains as many multiple bonded carbons as possible

ACS **52.** Which skeletal structure is the same molecule as the condensed structure $CH_3(CH_2)_4CH(CH_3)COOH$?

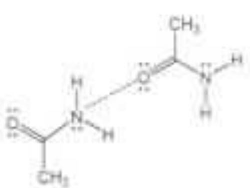
(A) 

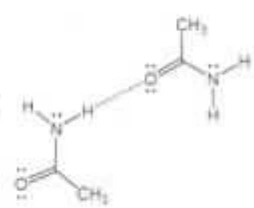
(B) 

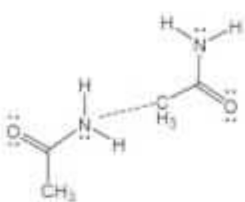
(C) 

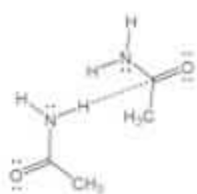
(D) 

ACS **53.** What best depicts hydrogen bonding between the two ethanamide molecules?

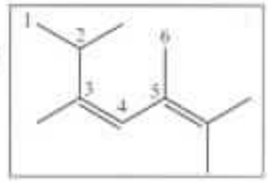
(A) 

(B) 

(C) 

(D) 

ACS **54.** Which C–C–C bond angle would be approximately 109.5°?



(A) C_1–C_2–C_3

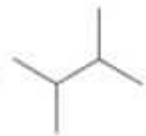
(B) C_2–C_3–C_4

(C) C_3–C_4–C_5

(D) C_4–C_5–C_6

ACS **55.** Which is an isomer of C_6H_{12}?

(A) 

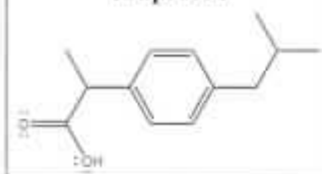
(B) 

(C) 

(D) 

ACS **56.** What is the molar mass of ibuprofen?

Ibuprofen



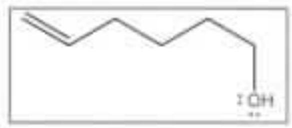
(A) 153 g·mol^{-1}

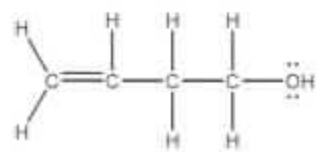
(B) 192 g·mol^{-1}

(C) 204 g·mol^{-1}

(D) 206 g·mol^{-1}

ACS **57.** What is another correct representation of this molecule?



(A) 

(B) $C_6H_{12}O$

(C) $CH_2{=}CH(CH_2)_4OH$

(D) $CH_2{=}CHCH_2CH_2CH_2CH_2CH_2OH$

ACS **58.** Which compound has the lowest boiling point?

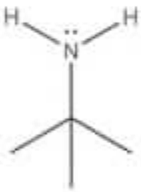
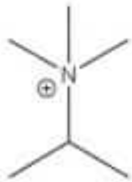
(A) $H_3C–CH_2–CH_2–CH_2–NH_2$

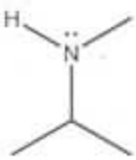
(B) $H_3C–CH_2–NH–CH_2–CH_3$

(C) $H_3C–CH_2–N–(CH_3)–CH_3$

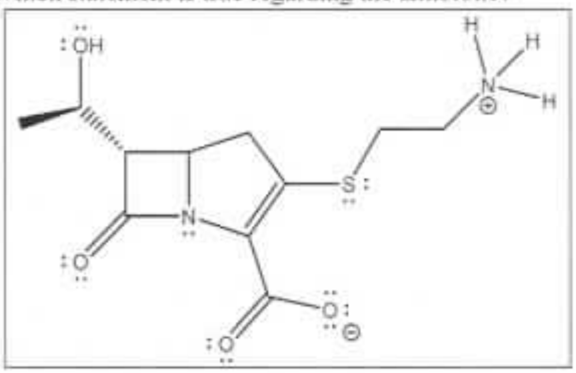
(D) All of the compounds have the same boiling point

ACS **59.** Which structure contains a tertiary amine?

(A)  (B) 

(C)  (D) 

ACS **60.** Which statement is true regarding the antibiotic?



(A) It will dissolve in nonpolar hexane.

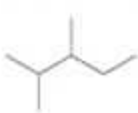
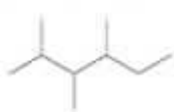
(B) It does not have polar functional groups.

(C) It is highly water soluble due to its ionic nature.

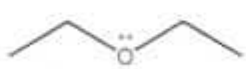
(D) It has too many polar functional groups to be soluble in water.

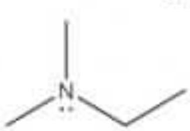
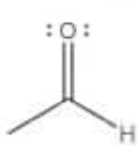
ACS **61.** Which molecule will exhibit the strongest London dispersion forces?

(A)  (B) 

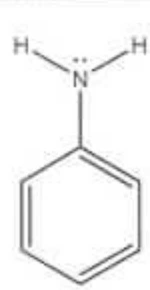
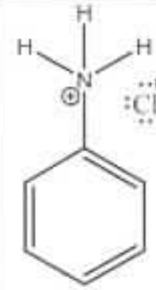
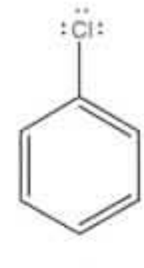
(C)  (D) 

ACS **62.** Intermolecular hydrogen bonding is found within a pure sample of which molecule?

(A)  (B) 

(C)  (D) 

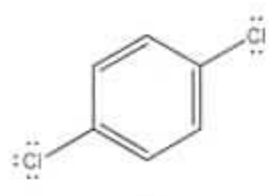
ACS **63.** Rank the compounds from least to most soluble in water.

		
A	**B**	**C**

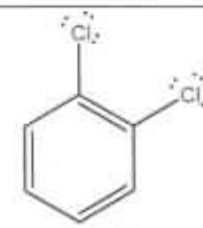
(A) A < B < C **(B)** A < C < B

(C) B < C < A **(D)** C < A < B

ACS **64.** Which statement is true?



X



Y

(A) Compound **X** is more polar than compound **Y**.

(B) Compound **Y** is more polar than compound **X**.

(C) Compounds **X** and **Y** have equal polarity.

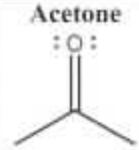
(D) Neither compound is polar.

ACS **65.** Which ether is least water soluble?

(A) dibutyl ether **(B)** diethyl ether

(C) dihexyl ether **(D)** dimethyl ether

ACS **66.** Why does acetone have a higher boiling point than butane?

Acetone	**Butane**
	
bp 56 °C	bp -1 °C

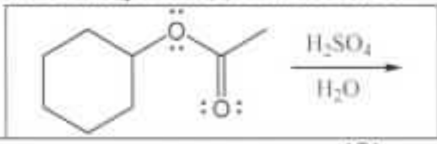
(A) Butane has a larger molecular mass.

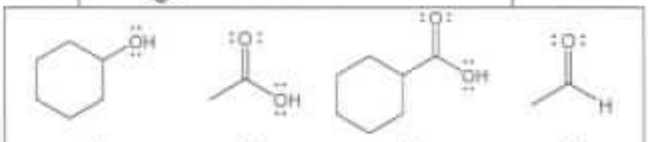
(B) Acetone has stronger intermolecular forces.

(C) Molecules of acetone form more hydrogen bonds than butane.

(D) Molecules of butane form more hydrogen bonds than acetone.

67. What is/are the product(s) of the reaction?



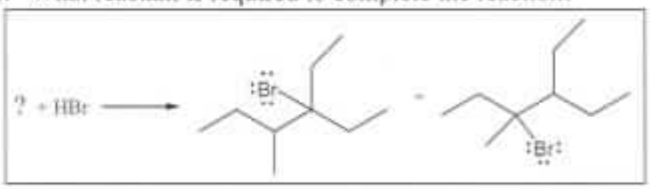


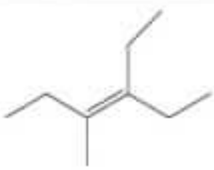
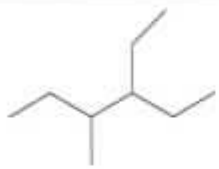
I II III IV

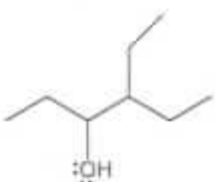
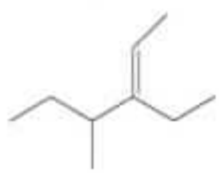
(A) I only **(B)** III only

(C) I and II only **(D)** III and IV only

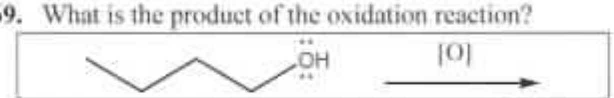
68. What reactant is required to complete the reaction?

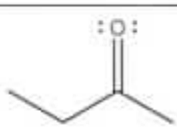
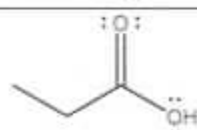


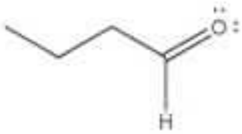
(A)  **(B)** 

(C)  **(D)** 

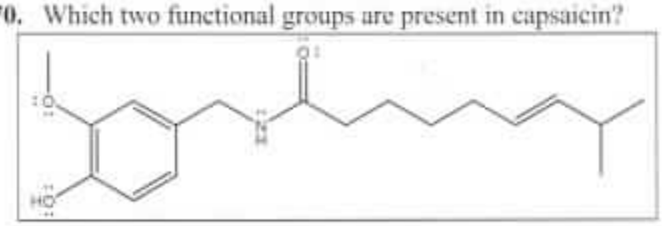
69. What is the product of the oxidation reaction?



(A)  **(B)** 

(C)  **(D)** 

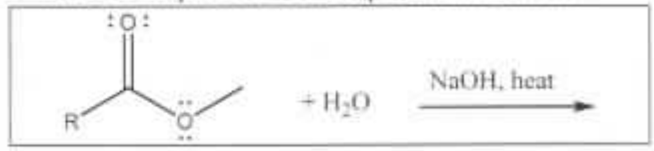
70. Which two functional groups are present in capsaicin?

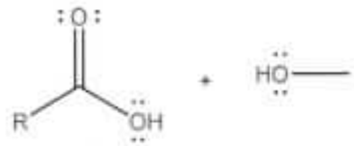


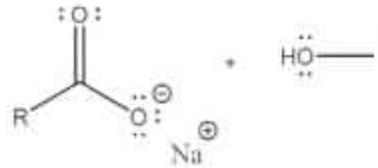
(A) amide and ether **(B)** amide and ester

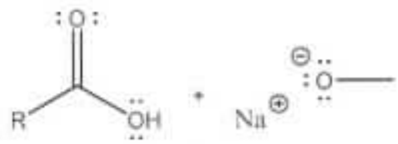
(C) amine and ether **(D)** amine and ester

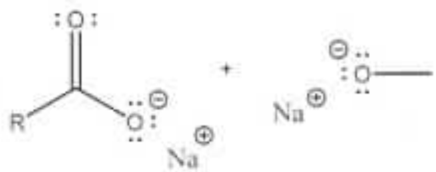
71. What are the products of the saponification reaction?



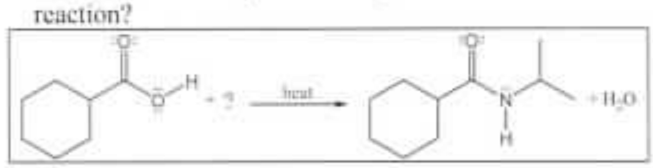
(A) 

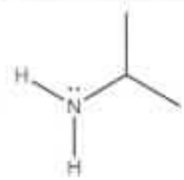
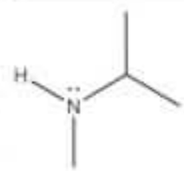
(B) 

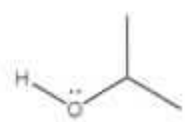
(C) 

(D) 

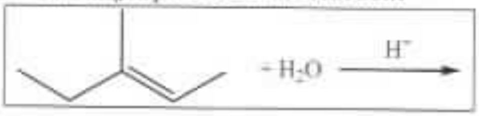
72. What reactant is required to complete the condensation reaction?

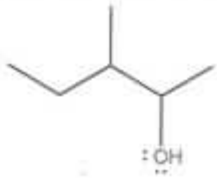


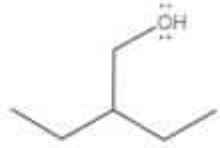
(A)  **(B)** 

(C)  **(D)** 

73. What is the major product of the reaction?



(A) 

(B) 

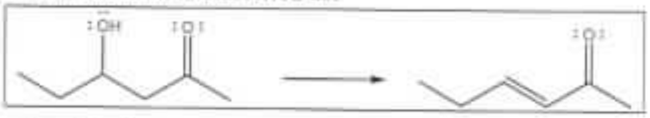
(C) 

(D) 

74. How is the reaction classified?

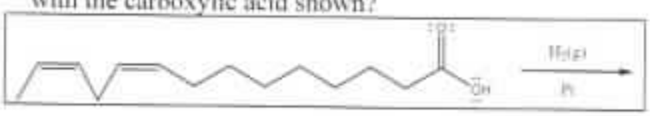
(A) dehydration
(B) hydration
(C) oxidation
(D) reduction

75. This reaction is classified as:



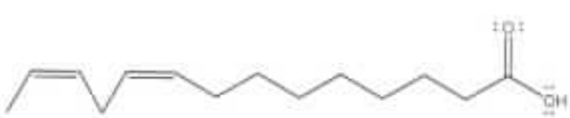
(A) dehydration
(B) hydration
(C) hydrolysis
(D) reduction

76. What product is formed when excess $H_2(g)$ and Pt react with the carboxylic acid shown?



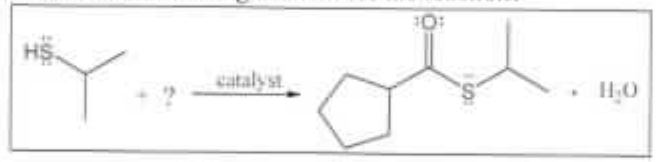
(A) 

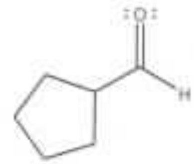
(B) 

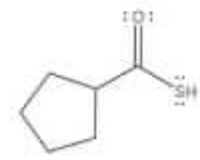
(C) 

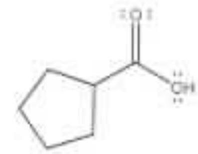
(D) 

77. What is the missing reactant for the reaction?



(A) 

(B) 

(C) 

(D) 

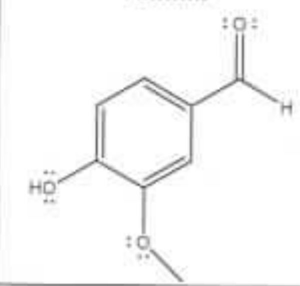
78. What is/are the possible product(s) of the reaction?

$$RCH_2NH_2 + H_2O \rightleftharpoons$$

(A) amide
(B) alkylammonium ion and hydroxide ion
(C) alkylammonium salt
(D) carboxylic acid and amide

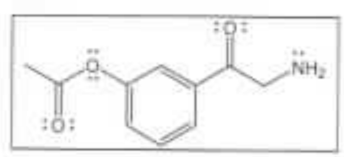
79. Which functional group is **not** present in vanillin?

Vanillin



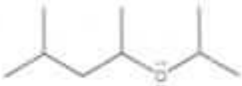
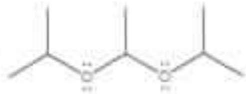
(A) aldehyde
(B) ether
(C) ketone
(D) phenol

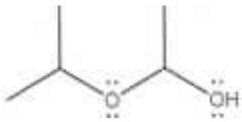
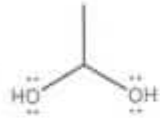
80. Which two functional groups are present in the molecule?



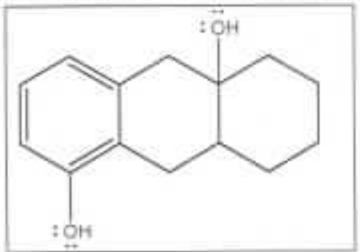
(A) amine and ether
(B) amine and ester
(C) amide and ether
(D) amide and ester

ACS 81. Which molecule is a hemiacetal?

(A)  (B) 

(C)  (D) 

ACS 82. What are the classifications of the two hydroxyl groups?

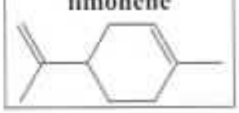


(A) 2° alcohol, 3°alcohol (B) 3° alcohol, 3°alcohol

(C) phenol, 2° alcohol (D) phenol, 3° alcohol

ACS 83. What term describes limonene, a component of essential oils in citrus fruits?

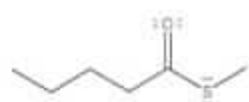
limonene



(A) aromatic (B) cycloalkane

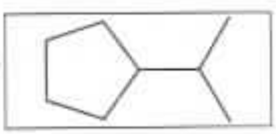
(C) cycloalkene (D) phenol

ACS 84. Which molecule is a thiol?

(A)  (B) 

(C)  (D) 

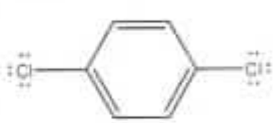
ACS 85. What is the correct name for the alkyl substituent attached to the cyclopentane ring?



(A) isobutyl (B) isopropyl

(C) n-propyl (D) t-butyl

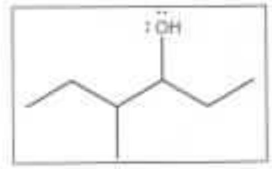
ACS 86. Which name best fits the structure?



(A) *o*-dichlorobenzene (B) dichlorobenzene

(C) *m*-dichlorobenzene (D) *p*-dichlorobenzene

ACS 87. Which name best fits the structure?



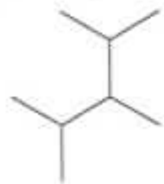
(A) 1-ethyl-2-methyl-butanol

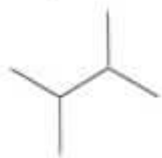
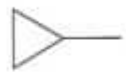
(B) 3-methyl-4-hexanol

(C) 4-hydroxy-3-methyl-hexanol

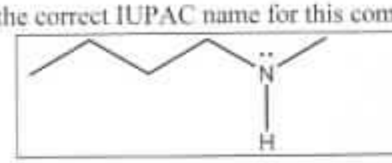
(D) 4-methyl-3-hexanol

ACS 88. Which compound has a parent butane chain?

(A)  (B) 

(C)  (D) 

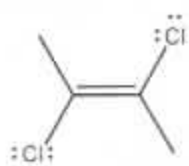
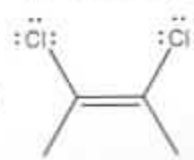
ACS 89. What is the correct IUPAC name for this compound?

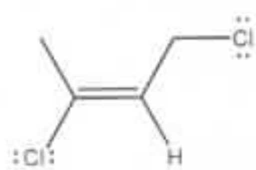
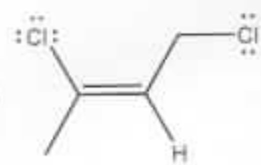


(A) 2-hexanamine (B) 2-pentanamine

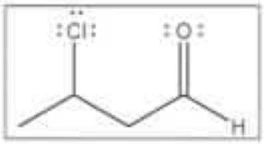
(C) N-butylmethanamine (D) N-methylbutanamine

ACS 90. Which structure depicts *trans*-2,3-dichloro-2-butene?

(A)  (B) 

(C)  (D) 

ACS **91.** What is the IUPAC name for this molecule?



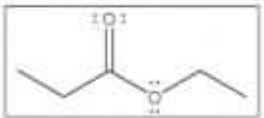
(A) chlorobutanoic acid

(B) 2-chloro-4-butanal

(C) 2-chloro-4-butanoic acid

(D) 3-chlorobutanal

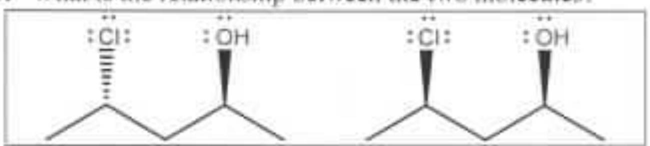
ACS **92.** Which name best fits the structure?



(A) ethyl ethanoate **(B)** ethyl propanoate

(C) pentanoate **(D)** propyl propanoate

ACS **93.** What is the relationship between the two molecules?



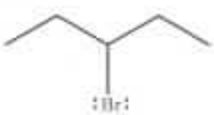
(A) They are diastereomers

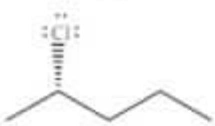
(B) They are enantiomers

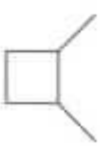
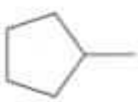
(C) They are structural/constitutional isomers

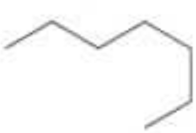
(D) They are the same molecule

ACS **94.** Which pair of molecules are structural/constitutional isomers?

(A)  and 

(B)  and 

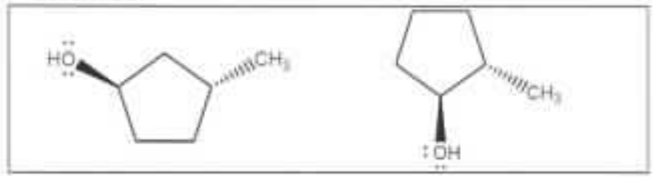
(C)  and 

(D)  and 

95. How many total structural/constitutional isomers of C_3H_8O are possible?

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5

ACS **96.** How is the relationship between the molecules best described?



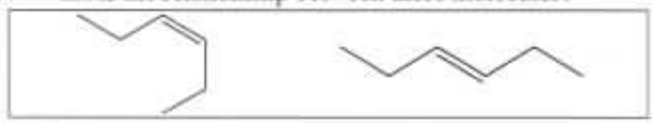
(A) They are diastereomers

(B) They are enantiomers

(C) They are structural/constitutional isomers

(D) They are the same molecule

ACS **97.** What is the relationship between these molecules?



(A) *cis-trans* isomers

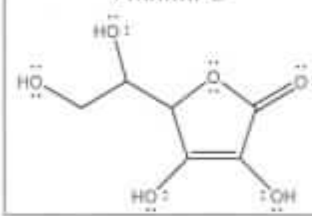
(B) diastereomers

(C) enantiomers

(D) structural/constitutional isomers

ACS **98.** How many chiral carbons are present in vitamin C?

Vitamin C



(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

ACS **99.** A racemic mixture consists of equal amounts of:

(A) a pair of enantiomers

(B) a pair of *cis-trans* isomers

(C) an acid and its conjugate base

(D) a strong acid and a strong base

ACS **100.** How is the relationship between cyclohexane and benzene best described?

(A) They are *cis-trans* isomers.

(B) They are structural/constitutional isomers.

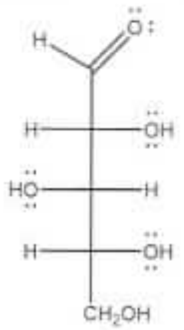
(C) They are the same molecule.

(D) They are not isomers.

End of Organic Chemistry Section

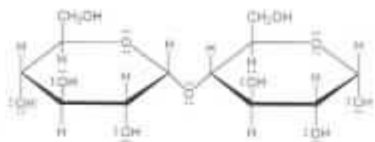
BIOCHEMISTRY SECTION

101. Which term can be used to describe this monosaccharide?



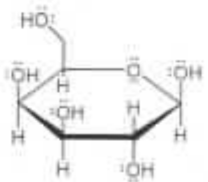
(A) alpha-configuration **(B)** D-configuration
(C) ketose **(D)** hexose

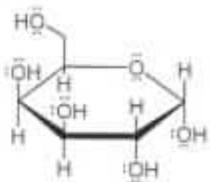
102. How is the glycosidic linkage classified?

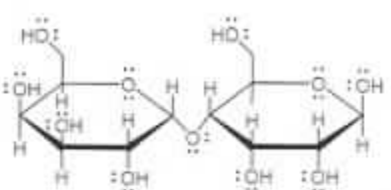


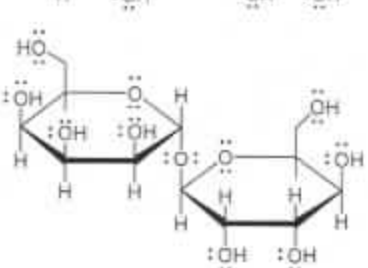
(A) α-1,4 **(B)** α-1,6 **(C)** β-1,4 **(D)** β-1,6

103. Which is NOT a reducing sugar?

(A)


(B)


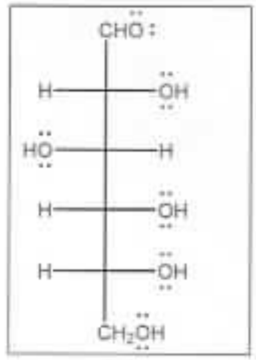
(C)


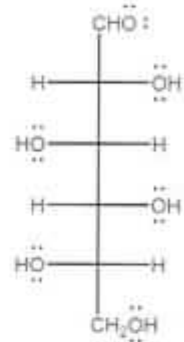
(D)


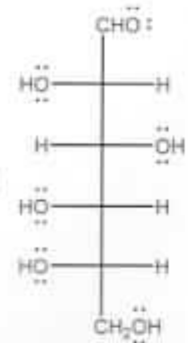
104. Which polymer of glucose is indigestible due to β linkages in its structure?

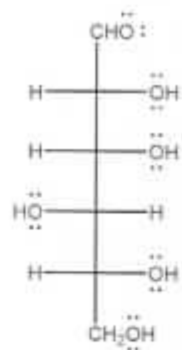
(A) amylopectin **(B)** amylose
(C) cellulose **(D)** glycogen

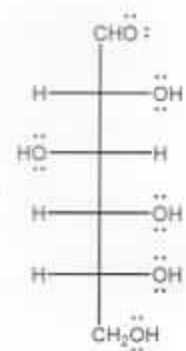
105. Which is the enantiomer to the structure?



(A)


(B)


(C)


(D)


106. Which molecule is permeable through the lipid bilayer?

(A) carbon dioxide **(B)** glycerol
(C) glucose **(D)** ATP

107. What is present in waxes, triacylglycerols, and glycolipids?

(A) fatty acid **(B)** glycerol
(C) glucose **(D)** phosphate

108. Which statements are TRUE regarding cholesterol?

I.	Cholesterol is a lipid that belongs to the steroid category.
II.	Cholesterol has no physiological role so consumption should be avoided.
III.	Cholesterol is hydrophobic due to its four-ring hydrocarbon structure.

(A) I and II **(B)** I and III

(C) II and III **(D)** I, II, and III

109. Coconut oil has recently gained popularity as an alternative cooking oil. Unlike vegetable oil, coconut oil is a solid at room temperature. Which statement describes the reason for the solid state of coconut oil?

(A) Coconut oil has a high percentage of trans fat.

(B) Coconut oil has a high percentage of cholesterol.

(C) Coconut oil has a high percentage of saturated fat.

(D) Coconut oil has a high percentage of short chain fatty acids.

110. Lipoproteins are spherical water-soluble lipid-protein complexes in which nonpolar lipids are surrounded by polar glycerophospholipids and proteins. What do lipoproteins help transport?

(A) DNA out of the nucleus

(B) polar molecules, like glucose, to different tissues

(C) nonpolar molecules like steroids to different tissues

(D) both polar and nonpolar molecules to different tissues

111. What are the products of protein hydrolysis?

(A) amino acids **(B)** fatty acids

(C) monosaccharides **(D)** nucleotides

112. Alpha helices are classified as what type of protein structure?

(A) primary **(B)** secondary

(C) tertiary **(D)** quaternary

113. Captopril, a hypertension drug, works by reversibly binding in the active site of the angiotensin-converting enzyme (ACE). What type of inhibition is this?

(A) competitive **(B)** mixed

(C) noncompetitive **(D)** allosteric

114. What interaction is found between nonpolar amino acid side chains in the tertiary structure of a protein?

(A) disulfide bond

(B) hydrogen bond

(C) ionic interaction

(D) hydrophobic interaction

115. Which amino acid side chain is most likely to be found on the exterior of a protein?

(A) $-CH_3$

(B) $-CH_2CH(CH_3)_2$

(C) $-CH_2CH_2SCH_3$

(D) $-CH_2CH_2CH_2CH_2NH_3^+$

116. Which important dietary element do proteins provide that other biomolecule classes do not?

(A) hydrogen **(B)** nitrogen

(C) oxygen **(D)** water

117. What is the physiological role of hemoglobin in the blood?

(A) store oxygen **(B)** transport oxygen

(C) generate oxygen **(D)** dissociate oxygen

118. Why must enzyme injections be maintained at a relatively constant pH?

(A) To slow the eventual enzyme denaturation.

(B) To protect the enzyme from stomach acid.

(C) To maintain physiological pH.

(D) To help reduce acid reflux.

119. Chymotrypsin catalyzes peptide bonds. To what class of molecules does chymotrypsin belong?

(A) enzymes **(B)** lipids

(C) nucleic acids **(D)** vitamins

120. Which pairing of vitamin and their function is correct?

(A) vitamin B; calcium fixation

(B) vitamin D; cholesterol lowering

(C) vitamin D; vision development

(D) vitamin K; blood clotting

121. What happens to an enzyme when it is denatured?

(A) It loses the tertiary structure.

(B) It loses the primary structure.

(C) It will decrease the temperature of the reaction.

(D) It will increase the activation energy of a reaction.

122. Most enzymes consist of a(n) _____.

(A) activator (B) coenzyme

(C) cofactor (D) protein

123. If a DNA strand has the sequence, 5'-ATTGCCAGA-3', what is the sequence of the complimentary strand?

(A) 3'-ATTGCCAGA-5' (B) 3'-GCCATTGAC-5'

(C) 3'-TAACGGTCT-5' (D) 3'-TAAGAATCA-5'

124. What is the structural similarity between RNA and DNA?

(A) total number of strands

(B) identity of nitrogenous bases

(C) identity of monosaccharide

(D) location of glycosidic bond

125. What is the role of tRNA in protein synthesis?

(A) provide protein sequence

(B) turn off gene translation

(C) bring amino acids to ribosome

(D) serve as integral part of ribosome structure

126. Which base pair is responsible for the H-bonding associations that hold DNA strands together?

(A) A – C (B) A – G (C) A – T (D) A – U

127. Which enzyme is responsible for unwinding DNA?

(A) helicase (B) ligase

(C) DNA polymerase (D) aminoacyl synthetase

128. Blood sugar concentrations are increased by _______.

(A) insulin (B) glucagon

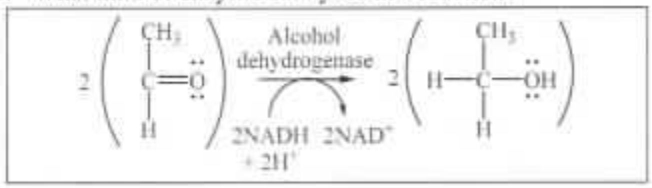
(C) cortisol (D) cholesterol

129. Which compound is formed from pyruvate under anaerobic conditions in humans?

(A) CoA (B) ascorbic acid

(C) ethanol (D) lactate

130. What kind of enzyme catalyzes this reaction?



(A) hydrolase (B) kinase

(C) ligase (D) oxidoreductase

131. Which is true for glycogenesis?

(A) It makes ATP.

(B) It stores glucose.

(C) It occurs in the mitochondria.

(D) It is the opposite of glycolysis.

132. ATP is a direct product of which biological process?

(A) beta-oxidation (B) glycolysis

(C) glycogenesis (D) gluconeogenesis

133. Which process(es) can be used to make glucose?

I.	glycogenolysis
II.	gluconeogenesis
III.	glycolysis

(A) II only (B) III only

(C) I and II (D) II and III

134. A saturated fatty acid is completely metabolized after 7 rounds of beta oxidation. What was the formula of the fatty acid?

(A) $C_7H_{14}O_2$ (B) $C_{14}H_{28}O_2$

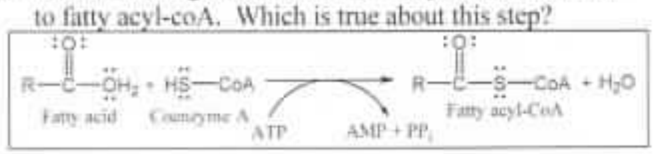
(C) $C_{16}H_{32}O_2$ (D) $C_{23}H_{42}O_2$

135. What type of plasma lipoproteins are known as "good" cholesterol?

(A) HDL (B) LDL

(C) VLDL (D) chylomicrons

136. Before entering beta oxidation, a fatty acid is activated to fatty acyl-coA. Which is true about this step?



(A) yields $FADH_2$

(B) requires energy

(C) also used in protein metabolism

(D) repeated for every two carbons of the fatty acid

137. When a triacylglycerol is used for energy which reaction is initially employed?

(A) combustion **(B)** condensation
(C) hydrolysis **(D)** isomerization

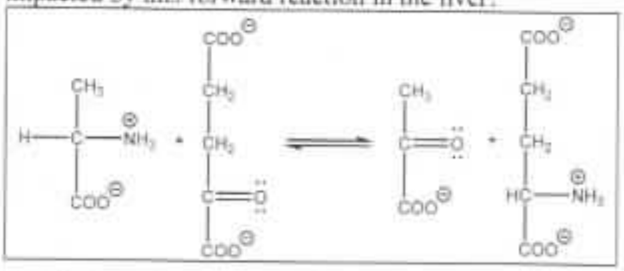
138. What is the starting material for the production of ketone bodies?

(A) acetone **(B)** acetyl CoA
(C) ethanol **(D)** oxaloacetate

139. Which is responsible for the first step in protein catabolism?

(A) kinase **(B)** lipase
(C) protease **(D)** transferase

140. The first step of which metabolic process is directly impacted by this forward reaction in the liver?



(A) alanine cycle **(B)** electron transport
(C) glycolysis **(D)** urea cycle

141. What product from beta oxidation is used in the citric acid cycle to produce NADH and $FADH_2$?

(A) acetyl CoA **(B)** ethanol
(C) fatty acids **(D)** glycerol

142. Which pathway will be upregulated when cells have excess glucose and sufficient energy?

(A) ketogenesis **(B)** glycogenesis
(C) gluconeogenesis **(D)** beta oxidation

143. Which metabolic process yields NAD^+ and FAD?

(A) glycolysis
(B) citric acid cycle
(C) fatty acid oxidation
(D) electron transport chain

144. In our body, which molecule cannot be broken down and cannot be stored as fat?

(A) carbohydrate **(B)** cholesterol
(C) protein **(D)** triacylgycerol

145. Rank triglycerides, blood sugar, and glycogen by the body's preference for energy.

(A) blood sugar > glycogen > triglycerides
(B) blood sugar > triglycerides > glycogen
(C) glycogen > triglycerides > blood sugar
(D) triglycerides > blood sugar > glycogen

146. How many ATP can be generated from the complete metabolic breakdown of two glucose molecules? Assume 1 NADH produces 3 ATP and 1 $FADH_2$ produces 2 ATP.

(A) 38 **(B)** 42 **(C)** 76 **(D)** 120

147. Which is not part of the electron transport chain?

(A) cyt *c* **(B)** $FADH_2$
(C) oxygen **(D)** oxaloacetate

148. The main function of the citric acid cycle is to directly produce:

(A) ATP
(B) acetyl-CoA
(C) the oxidized forms of citric acid
(D) the reduced forms of NADH and FAD

149. Some of the energy conserved in ATP and subsequently used in muscle contraction is an example of what type of bioenergetics transformation?

(A) heat energy to electrical energy
(B) chemical energy to heat energy
(C) chemical energy to mechanical energy
(D) mechanical energy to electrical energy

150. Which conversion is not possible in a human cell?

(A) ethanol → acetaldehyde
(B) pyruvate → alanine
(C) alanine → pyruvate
(D) ethanol → pyruvate

END OF TEST

ABBREVIATIONS AND SYMBOLS

amount of substance	n	enzyme	E	liter	L
aqueous	aq	equilibrium constant	K	measure of pressure	mmHg
atmosphere	atm	formula molar mass	M	milli- (10^{-3})	m
Avogadro constant	N_A	gram	g	molar	M
Celsius temperature	°C	joule	J	mole	mol
centi- (10^{-2})	c	kelvin	K	pressure	P
energy of activation	E_a	kilo- (10^3)	k	temperature, K	T
				volume	V

gas constant: $R = 0.0821$ L·atm·mol^{-1}·K^{-1} **Note:** Notation such as g·mol^{-1} is read as "grams per mole."

PERIODIC TABLE OF THE ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 **H** 1.008																	2 **He** 4.003
3 **Li** 6.941	4 **Be** 9.012											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
11 **Na** 22.99	12 **Mg** 24.31											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.07	17 **Cl** 35.45	18 **Ar** 39.95
19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.88	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.85	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.39	31 **Ga** 69.72	32 **Ge** 72.61	33 **As** 74.92	34 **Se** 78.96	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.94	43 **Tc**	44 **Ru** 101.1	45 **Rh** 102.9	46 **Pd** 106.4	47 **Ag** 107.9	48 **Cd** 112.4	49 **In** 114.8	50 **Sn** 118.7	51 **Sb** 121.8	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.3
55 **Cs** 132.9	56 **Ba** 137.3	57 **La** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.8	75 **Re** 186.2	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209.0	84 **Po**	85 **At**	86 **Rn**
87 **Fr**	88 **Ra**	89 **Ac**	104 **Rf**	105 **Db**	106 **Sg**	107 **Bh**	108 **Hs**	109 **Mt**	110 **Ds**	111 **Rg**	112 **Cn**	113 **Nh**	114 **Fl**	115 **Mc**	116 **Lv**	117 **Ts**	118 **Og**

58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm**	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3	65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
90 **Th** 232.0	91 **Pa** 231.0	92 **U** 238.0	93 **Np**	94 **Pu**	95 **Am**	96 **Cm**	97 **Bk**	98 **Cf**	99 **Es**	100 **Fm**	101 **Md**	102 **No**	103 **Lr**